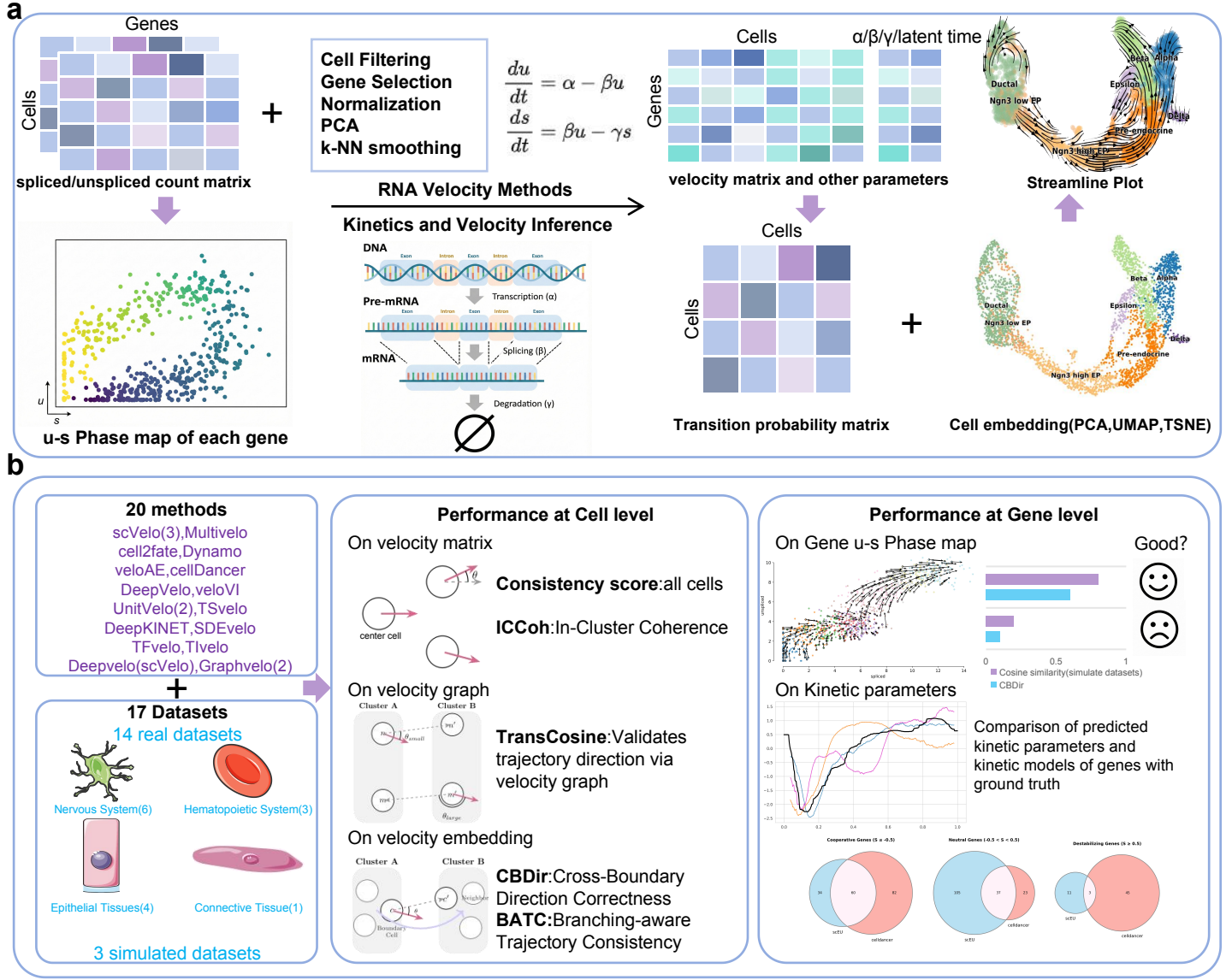

a
Genes
Cells
spliced/unspliced count matrix
+
Cell Filtering
Gene Selection
Normalization
PCA
k-NN smoothing
$\frac{du}{dt} = \alpha - \beta u$
$\frac{ds}{dt} = \beta u - \gamma s$
RNA Velocity Methods
Kinetics and Velocity Inference
DNA
Transcription (α)
Pre-mRNA
Splicing (β)
mRNA
Degradation (γ)
u
s
u-s Phase map of each gene
Cells
Genes
α/β/γ/latent time
velocity matrix and other parameters
Ductal
Ngn3 low EP
Ngn3 high EP
Pre-endocrine
Beta
Alpha
Epsilon
Delta
Streamline Plot
Cells
Cells
Transition probability matrix
+
Cell embedding(PCA,UMAP,TSNE)
b
20 methods
scVelo(3),Multivelo
cell2fate,Dynamo
veloAE,cellDancer
DeepVelo,veloVI
UnitVelo(2),TSvelo
DeepKINET,SDEvelo
TFvelo,TIvelo
Deepvelo(scVelo),Graphvelo(2)
+
17 Datasets
14 real datasets
Nervous System(6)
Hematopoietic System(3)
Epithelial Tissues(4)
Connective Tissue(1)
3 simulated datasets
Performance at Cell level
On velocity matrix
center cell
Consistency score:all cells
ICCoh:In-Cluster Coherence
On velocity graph
Cluster A
Cluster B
TransCosine:Validates trajectory direction via velocity graph
On velocity embedding
Cluster A
Cluster B
Boundary Cell
Neighbor
CBDir:Cross-Boundary Direction Correctness
BATC:Branching-aware Trajectory Consistency
Performance at Gene level
On Gene u-s Phase map
Good?
unspliced
spliced
Cosine similarity(simulate datasets)
CBDir
On Kinetic parameters
Comparison of predicted kinetic parameters and kinetic models of genes with ground truth
Cooperative Genes (S ≤ -0.5)
Neutral Genes (-0.5 < S < 0.5)
Destabilizing Genes (S ≥ 0.5)
scEU
celldancer